1. AgriAgent Week 2 Evidence Dossier

Review date: 11 October 2026
Scope: repository implementation and checks available in this checkout
Status: in progress; this report does not certify field readiness.

1.1. Project summary

AgriAgent is a research prototype that processes field images, creates candidate vegetation masks, runs agronomy reasoning, and prepares spray-map artifacts for review. A deterministic gate owns spray safety. Local Clef Flash can assist ambiguity triage but cannot authorize spraying.

The intended models are local `Cloudflare/clef-flash`, local `Qwen/Qwen3.5-4B`, and cloud-only Groq `qwen/qwen3.8-27b`. Local model execution has not been verified in this environment. Tests using `mock_mode=True` verify software contracts, not model quality.

1.2. Evidence matrix

Area	Status	Evidence	Remaining work
Vision pipeline	Partial	`src/workflow.py`, `src/vision/grounding_dino.py`, `src/vision/sam2_wrapper.py`; mock/unit workflow tests.	Run with pinned real weights and independently annotated field images; review errors and calibration.
Spray safety	Repository contract verified	`src/workflow.py`, `app/components/pipeline.py`, `tests/test_spray_export_safety.py`; six safety regressions pass, including crop rejection, grass acceptance, suppression, raw-box rejection, and mask-derived geometry.	Review every label policy and physical suppression threshold with an agronomist and hardware team; validate GPS calibration and actuators.
Local Clef	Implemented; runtime unverified	`src/agents/clef_flash_router.py` uses local cache-only loading and escalation on load failure.	Install model/runtime, verify image processor compatibility, memory, triage quality, and offline startup on target hardware.
Local Qwen3.5-4B	Partial	`src/agronomy/providers/local_qwen.py`, `src/agronomy/vlm_critic.py`, provider mock tests.	Confirm serving stack compatibility and run a live local diagnosis with recorded model revision and device.
Groq Qwen3.8-27B	Partial	`src/agronomy/providers/hosted_groq.py` defaults to requested model; provider mock/config tests.	Verify model availability and response schema using an authorized live API key.
Evaluation	Mock and manifest-based code	`src/evaluation/ablation_runner.py`, `data/manifests/eval_manifest_v1.json`; mock-mode runs do not measure model performance.	Review labels/splits and rerun real models; retain provenance, revisions, device, and output metadata.
CI	Local Python 3.11 checks pass; GitHub run failed on prior commit	CI run 38150770034 failed mypy on both Python 3.10 and 3.11 before later test steps. In the local Python 3.11 CI-equivalent environment, compileall, Ruff check/format, mypy (27 source files), 175 unit tests, and 35 E2E tests now pass.	Commit and push the typing/runtime fixes, confirm a fresh Python 3.10/3.11 Actions run, and run the Streamlit smoke test; local socket binding was blocked by sandbox permissions.

Documentation	Mostly updated; final review pending	`README.md`, `docs/models_and_runtime.md`, `docs/cleanup_execution_sprint.md`, Week 1 sources, this dossier; updated PDFs regenerated with Typst.	Review active documentation and generated PDFs after the final CI matrix.

1.3. Safety contract

Grounding DINO proposals may prompt SAM 2 but are not spray instructions. The workflow filters mask labels, forms an approved spray-map artifact, and applies the deterministic infestation gate. A suppressed result must carry no approved zones. GeoJSON must read only those artifact zones; it must not reconstruct polygons from raw detector boxes.

The current coordinates are illustrative placeholders. This repository has no verified GPS calibration, nozzle hardware integration, vehicle interlocks, or field safety case. No generated GeoJSON is approved for real equipment.

1.4. Checks required for release

Run from the repository root:

```
python -m compileall src/ app/ tests/
ruff check .
ruff format --check .
mypy --config-file pyproject.toml src/
PYTHONPATH=. python -m unittest discover tests -v
PYTHONPATH=. python tests/e2e/runner.py --all
```

CI additionally runs a headless Streamlit health check. The local checks above passed in a Python 3.11 environment with the project and CI dependencies installed. The only completed GitHub run (38150770034, for the preceding commit) failed mypy on both matrix versions; this local fix set has not yet been pushed or rechecked by GitHub. The Streamlit smoke test remains unverified because the local sandbox denied binding a localhost socket.

1.5. Limitations

- The current environment did not expose a working NVIDIA driver; GPU use and live inference are unverified.
- Approximate memory planning is in `docs/models_and_runtime.md`; it is not a measured benchmark.
- Unit and end-to-end tests can use synthetic inputs and mocked model responses.
- The prototype is not field-ready. Model quality, agronomic correctness, calibration, regulatory review, network/service operation, and physical actuation all need separate validation.

For the detailed task breakdown and acceptance criteria, see `docs/cleanup_execution_sprint.md`.